

Data Cleaning & Preprocessing using Python

Project Overview

This project demonstrates a basic data cleaning pipeline using Python and Pandas. The objective was to transform a raw sales dataset containing missing values and inconsistencies into a clean, analysis-ready dataset.

Input

File: sample_sales.csv

The raw dataset contained:

- Missing names
- Missing ages
- Missing email addresses
- Missing sales values
- Inconsistent records

Dataset Size:

- 20 rows
- 6 columns

Columns:

- Name
 - Age
 - Email
 - Sales
 - City
 - Department
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Processing Performed

The Python script performs the following cleaning operations:

- ✓ Handles missing values
- ✓ Replaces missing names with "Unknown"
- ✓ Replaces missing email addresses with "Unknown"
- ✓ Fills missing age values using appropriate statistical replacement
- ✓ Fills missing sales values

- ✓ Removes invalid or unusable records
 - ✓ Produces a structured and analysis-ready dataset
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Output

File: sample_sales_cleaned.csv

Result:

- Cleaned dataset
- No critical missing values
- Consistent formatting
- Ready for data analysis and visualization

Dataset Size:

- 19 rows
 - 6 columns
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Technologies Used

- Python
 - Pandas
 - CSV File Handling
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Key Learning Outcomes

- Data Cleaning Techniques
 - Missing Value Treatment
 - Data Preprocessing
 - Working with Pandas DataFrames
 - Preparing datasets for Data Analysis and Machine Learning
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GitHub: <https://github.com/silverbullet-ai>